

PAMAM type dendritic electrolytes for lithium ion battery applications

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ABSTRACT

Series of boron-complexed ion conducting dendritic electrolytes were prepared and suggested as ‘cobweb’ model. The hydroxyl functionalized Jeffamine® cored polyamidoamine (PAMAM) type dendrimers with various core size and generation were synthesized and their boron-cross linked form was obtained by using $\text{BH}_3\text{--THF}$ complex. During cross linking of the dendrimers, lithium trifluoromethane-sulfonate ($\text{CF}_3\text{SO}_3\text{Li}$) salt was added as dopants with different stoichiometric ratios. The structure and interactions of boron-cross linked dendritic electrolytes were investigated by using FT-IR, NMR and Thermogravimetric analysis (TGA). The ionic conductivities of these novel boron-cross linked dendritic electrolytes were studied by dielectric-impedance spectroscopy. Ionic conductivity of the dendritic electrolytes followed Vogel–Tammann–Fulcher (VTF) relationships. The results showed that the highest conductivity of $1.6 \times 10^{-4} \text{ S/cm}$ was obtained at 100°C for electrolytes T5000(6-OH)–20.

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1. Introduction

Dendrimers are a unique class of macromolecules with uniform branching repeat unit due to their monodispersity in size, three-dimensional structure, and various functionalities [1]. A lot of studies have been done on the synthesis, characterization and modification of novel dendrimers [2]. Among all the dendrimers, PAMAM dendrimers have been most studied due its various applications such as the biomedical science, the environmental science, and the material science [3]. Still the needs for new applications are desired in spite of the great development in dendrimer technology.

The most important component of the lithium polymer battery is the electrolyte. The proper choice of the electrolyte is necessary for application where this material should have high ionic conductivity, good mechanical properties and compatibility [4,5]. Liquid electrolytes have higher conductivity, but solid electrolytes are preferred because the former can leak, but the latter have better film formation properties [6,7]. Ion-conducting polymers are solutions of salts in polymers in which a macroscopically solid state is achieved by entanglement or cross-linking. The polymer electrolytes are generally composites of a polyethylene oxide and a salt such as LiClO_4 , LiAsF_6 , or LiCF_3SO_3 [5,8–10]. It is generally believed that ionic conduction is a property of the amorphous phase and that ion association, ion–ethylene oxide interactions, and local relaxations strongly influence the ionic mobility [10].

Dendrimers containing electronegative heteroatoms in the chain branches, such as etheric oxygens, have been used as hosts, for use

in ion conductive polymer electrolytes. The amorphous nature of dendritic organic molecules makes them attractive as the host for electrolytes. In addition, these branched polymers contain a large number of chain ends, which should result in a high free volume and low glass transition temperatures [11–13]. Relatively high ambient conductivity is the important property of such electrolytes.

Herein, we have synthesized hydroxyl functional group containing Jeffamine® cored PAMAM type dendrimers and their boron-cross linked form by using $\text{BH}_3\text{--THF}$ complex for lithium ion battery applications. Then dendritic electrolytes were successfully prepared by doping of the boron-cross linked dendrimers with $\text{CF}_3\text{SO}_3\text{Li}$ at several stoichiometric ratios. The materials were characterized by using Fourier transform infrared spectroscopy (FT-IR), nuclear magnetic resonance ($^1\text{H-NMR}$), thermogravimetry (TG). The materials are thermally stable up to 200°C . The ionic conductivity of these novel dendritic electrolytes was studied by dielectric-impedance spectroscopy. Li-ion conductivity of these dendritic electrolytes depends on the length of the core and generation of dendrimers as well as the doping ratio. The dendritic electrolyte T5000(6-OH) B-20 illustrated an ionic conductivity of $\sim 5.6 \times 10^{-7} \text{ S/cm}$ at 20°C and $\sim 1.55 \times 10^{-4} \text{ S/cm}$ at 100°C . At higher doping ratios, Vogel–Tammann–Fulcher (VTF) behavior of conductivity implies the coupling of the charge carriers with the segmental motion of the dendrimer chains.

2. Experimental

2.1. Materials

Poly(propylene oxide) based Jeffamine® T series products (MW = 440, 3000 and 5000 g/mol) were gifts from HUNTSMAN International,

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LLC (Turkey). Methyl acrylate (99.5%) and ethylenediamine (99.8%) were obtained from Merck. Borane–tetrahydrofuran complex (BH_3/THF , 1.0 M solution in THF) and lithium trifluoromethane sulfonate ($\text{CF}_3\text{SO}_3\text{Li}$, 97%) were obtained from Alfa Aesar and stored in the glove box. All the other chemicals were used as received without further purification.

2.2. Synthesis of Jeffamine cored PAMAM dendrimers (JAPD)

Divergent synthesis of the Jeffamine cored (with different MWs) amine-terminated PAMAM dendrimers were synthesized according to our previous study [14]. Dendrimer synthesis was carried out by initial Michael addition of methanolic solution of Jeffamine T series with excess methyl acrylate (1:10 molar ratio). The reaction mixture was stirred for 3 days at room temperature. The excess methylacrylate was removed under vacuum at 40–50 °C temperature to afford the ester-functionalized derivatives. The reaction mixture was next submitted to the reaction sequence leading to the next generation PAMAM dendrimers, consisting of the exhaustive amidation of the ester functionalized Jeffamines to ethylenediamine (1:30 molar ratio), followed by Michael addition of the resulting amine with methyl acrylate (20 equiv of ester terminated dendrimer). Excess reagents were removed under vacuum at 50–60 °C temperature. Repetition of this two-step procedure ultimately leads to the next generation PAMAM dendrimers (2, 3, 4 and 5). The dendrimers, isolated in ~80–90% yield,

were gummy in nature. The syntheses of the dendrimers were confirmed through ^1H NMR and FTIR spectroscopy.

2.3. Preparation of hydroxyl-terminated dendrimers

Jeffamines and its core PAMAM dendrimers were dissolved in DMF and then chlorohydrin (1:1 mol/mol NH_2 group) was added drop wise under vigorous stirring at room temperature for 30 min. The mixture solution was heated at 60 °C for 12 h under vigorous stirring. After reaction was completed, the solvent was removed by vacuum distillation and the residue was dried in vacuum oven at 40 °C for 24 h.

2.4. Synthesis of boron containing crosslinked dendritic electrolytes

The boron containing crosslinked dendrimers were prepared by using BH_3/THF . Firstly the prepared hydroxyl-terminated dendrimers were dissolved in DMF, and then BH_3/THF was added slowly to the prepared mixed solution. The resulting solution was stirred for 12–18 h at 25–50 °C. Because the crosslinked dendrimers are not soluble in common organic solvents, the dendritic electrolytes were prepared by adding of appropriate amounts of lithium trifluoromethane-sulfonate ($\text{CF}_3\text{SO}_3\text{Li}$, 97%) into the reaction medium. Weighed amounts of dendrimers and salts were adjusted for the different levels of salt concentration corresponding to the ratios $[\text{PO}]:[\text{Li}] = 20$ and 40. The synthesized crosslinked dendrimers were precipitated

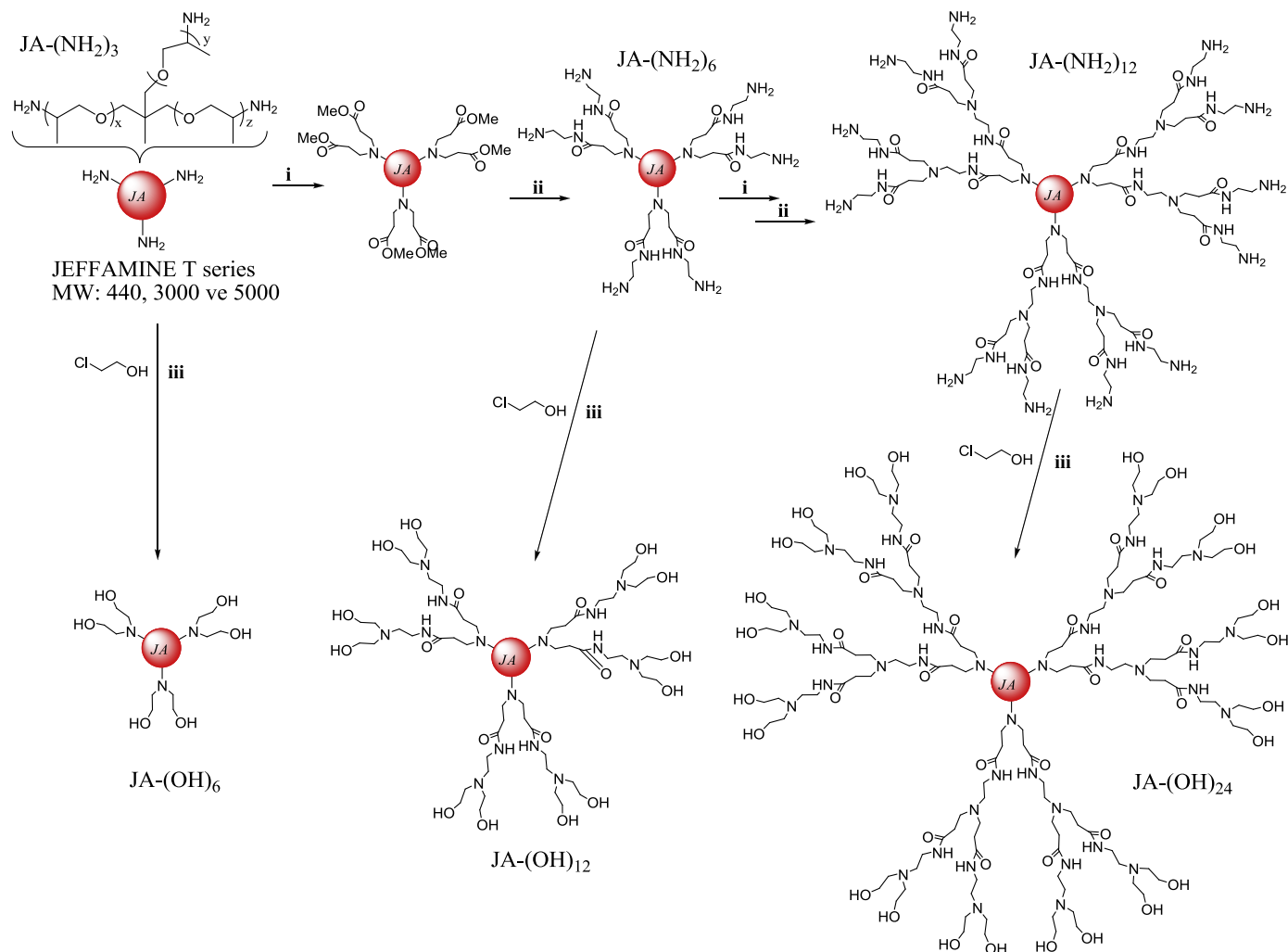


Fig. 1. The synthesis of hydroxyl terminated Jeffamine core PAMAM dendrimers.

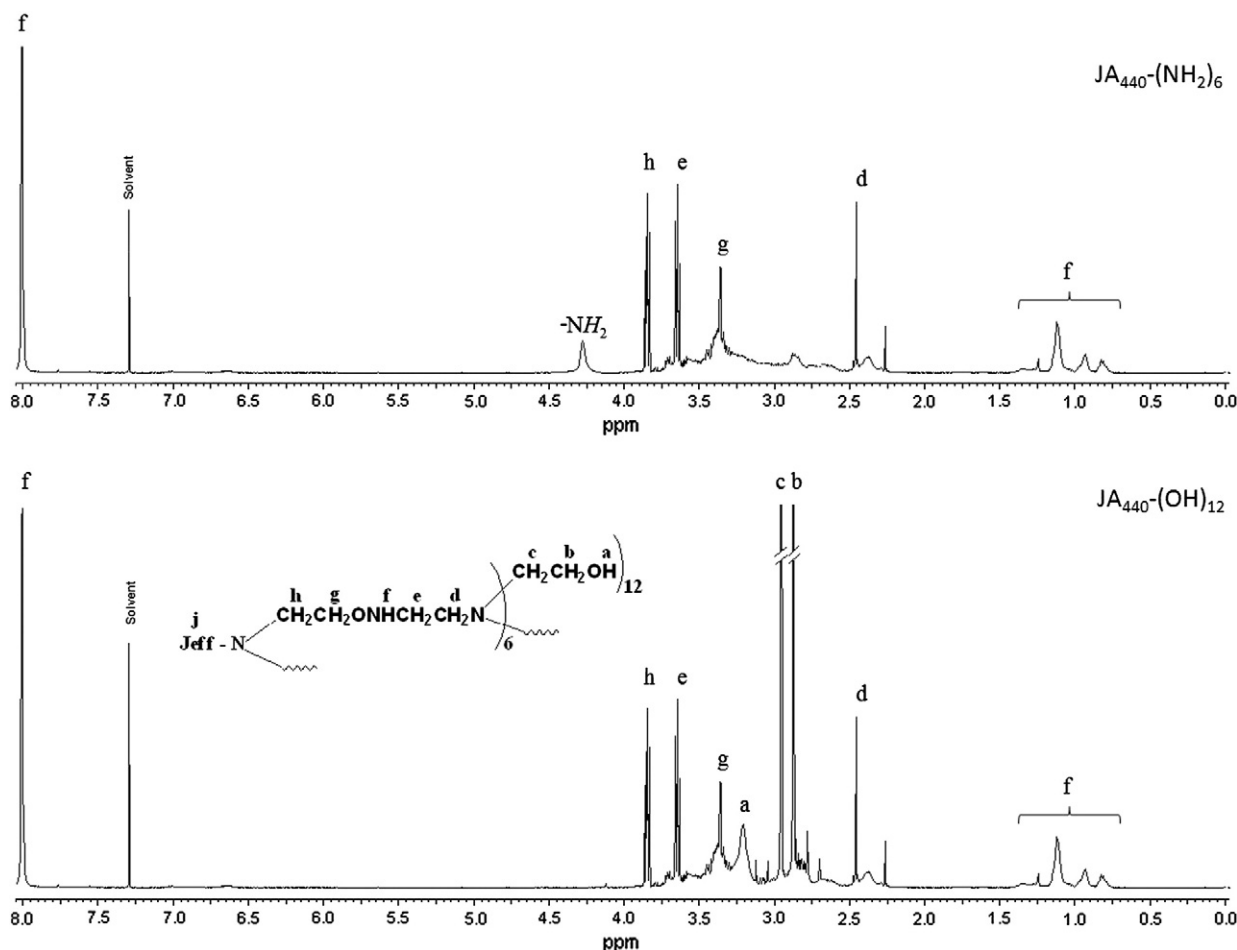


Fig. 2. ^1H NMR spectra of $\text{JA}_{440}-(\text{NH}_2)_6$ and $\text{JA}_{440}-(\text{OH})_{12}$.

by using diethylether. Then crosslinked dendrimers were dried in a vacuum oven at 50°C for several days to remove residual solvent.

Boron containing crosslinked dendritic electrolytes were produced and abbreviated as $\text{TA}(\text{XOH})\text{B-Li}$. The synthesis and the handling of the dendritic electrolytes were made under protective argon atmosphere. Finally the transparent, light yellow and sticky dendritic electrolytes were obtained and stored in a glove box (moisture content below 1 ppm).

2.5. Characterization

NMR spectra were recorded in CDCl_3 using a Bruker 400 MHz spectrometer.

Prior to Fourier transform infrared (FT-IR) spectra measurements, samples were dried under vacuum and stored in a glove box. The IR spectra ($4000\text{--}400\text{ cm}^{-1}$, resolution 4 cm^{-1}) were recorded at room temperature using a Bruker Alpha-P in ATR system coupled to a computer.

Thermal stabilities of the obtained dendritic electrolytes were examined by thermogravimetry analysis (TGA) with a Perkin Elmer Pyris 1. The samples ($\sim 10\text{ mg}$) were heated from room temperature to 700°C under nitrogen atmosphere at a heating rate of $10^\circ\text{C min}^{-1}$.

2.6. Conductivity measurements

Ionic conductivities of the electrolyte samples with different core size and generation of dendrimers were measured by the ac impedance analyzer in the frequency range from 0.1 Hz to 3 MHz as a

function of temperature. The samples with a diameter of 10 mm and a thickness of approximately 0.5 mm were sandwiched between two Pt electrodes and their conductivities were measured at 10 K intervals under a dry nitrogen atmosphere. The ion dynamics in the dendritic electrolytes and the frequency dependence of the conductivity at various temperatures have been analyzed for all samples.

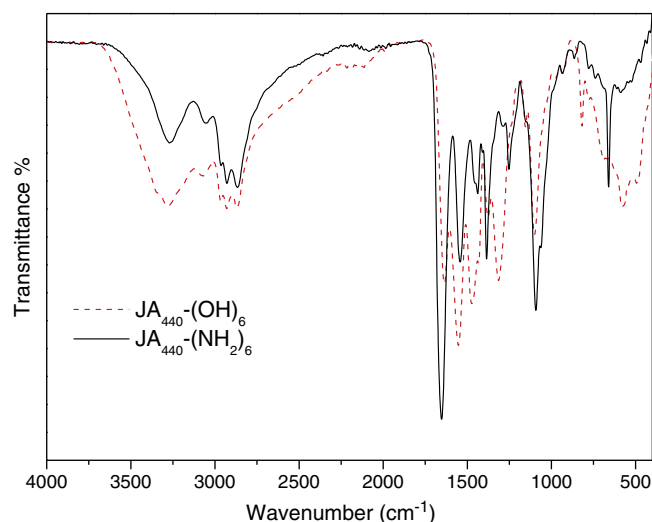


Fig. 3. FTIR spectra of $\text{JA}_{440}-(\text{NH}_2)_6$ and $\text{JA}_{440}-(\text{OH})_6$.

Table 1
FT-IR bands for JA₄₄₀-(NH₂)₆ and JA₄₄₀-(OH)₆ dendrimers.

Assignment	JA ₄₄₀ -(NH ₂) ₆ ^a	JA ₄₄₀ -(OH) ₆ ^a
Amide A	3284	3425
Amide B	3075	3078
CH ₂ antisymmetric stretching	2936	2940
CH ₂ symmetric stretching	2865	2866
OH	–	3500
Amide I	1644	1645
Amide II	1556	1555

^a Band positions (cm⁻¹).

3. Results and discussion

3.1. Preparation of hydroxyl terminated Jeffamine cored PAMAM dendrimers

The Jeffamine cored PAMAM dendrimers were synthesized via divergent approach as shown in Fig. 1. The synthesized dendrimers were characterized by ¹H NMR and FTIR spectral analysis and the results agreed with that reported in the literature [15]. The NH₂ functionalized dendrimers were reacted with 2-chloroethanol in DMF to obtain OH functionalized dendrimers [16].

The ¹H NMR spectra of the synthesized JA₄₄₀-(NH₂)₆ and JA₄₄₀-(OH)₁₂ dendrimer are shown in Fig. 2. The peaks can be assigned as Fig. 2; the peak at 3.24 ppm is from the proton of –OH.

Fig. 3 shows a comparison the FTIR spectra of JA₄₄₀-(NH₂)₆ and JA₄₄₀-(OH)₁₂ between 4000 and 500 cm⁻¹. Surface hydroxylation of NH₂ functionalized dendrimers led to increase band at 3350 cm⁻¹. The band positions and their assignments are listed in Table 1. The band at 3500 cm⁻¹ is due to the –OH group for hydroxyl-terminated dendrimer, the band positions of amides I and II (1644 and 1555 cm⁻¹) of dendrimer seem to be invariant, while the amides A and B shift to higher regions (3425 and 3078 cm⁻¹) after hydroxyl modification. All of these NMR and FT-IR spectra changes indicate the formation of hydroxyl-terminated dendrimer.

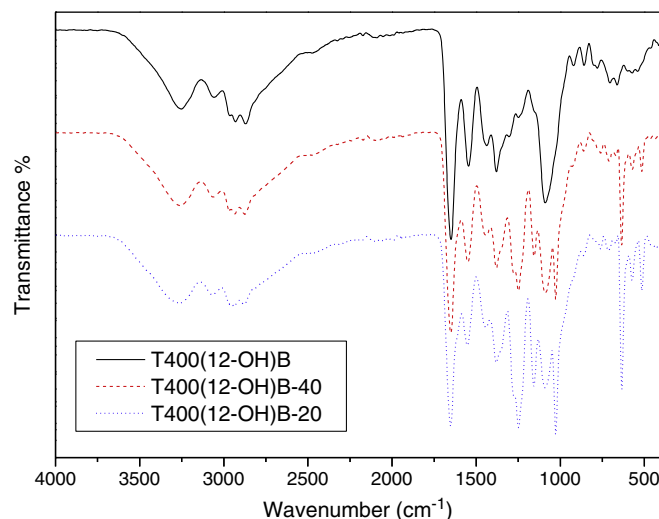


Fig. 5. Comparison of FTIR spectra of T400(12OH)B-Li (Li = 20 and 40).

3.2. Preparation of boron containing crosslinked dendritic electrolytes

The *cobweb model* boron crosslinked form of dendrimers was prepared at 50 °C within 24 h in DMF by reacting a BH₃/THF complex with the dendritic support and the ion conductive dendritic electrolytes were obtained by doping Li-salt (Fig. 4). The reaction of dendrimer with BH₃/THF complex was confirmed by FT-IR. These materials were characterized by TGA and the ionic conductivity was studied by dielectric-impedance spectroscopy.

Comparison of FTIR spectra of T400(12OH)B-Li (Li = 20 and 40) with different doping ratios is shown in Fig. 5. Boron containing dendritic electrolytes show weak peaks at 665 cm⁻¹ due to B–O units [16,17]. The absorption bands at ~1029 and ~1350 were assigned to the vibration of B–O–C bond and to the antisymmetric stretching vibration of B–O, respectively, which confirms the reaction of BH₃ with hydroxyl units of dendrimer [18]. After increasing the doping ratio, there is a broadening

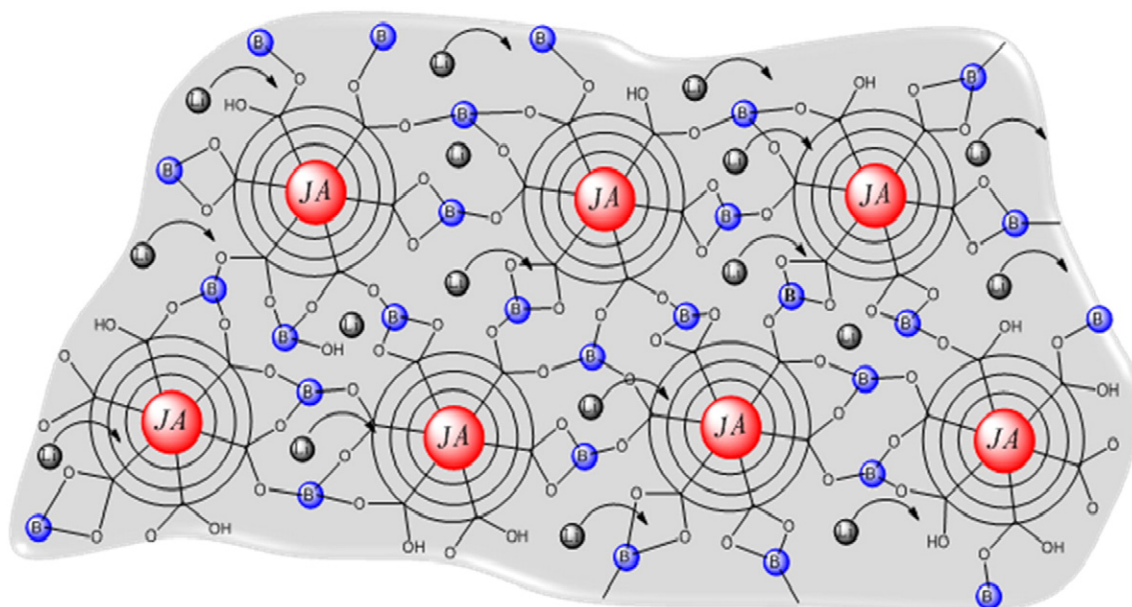


Fig. 4. Illustration of proposed structure of *cobweb model* boron-complexed dendritic electrolytes.

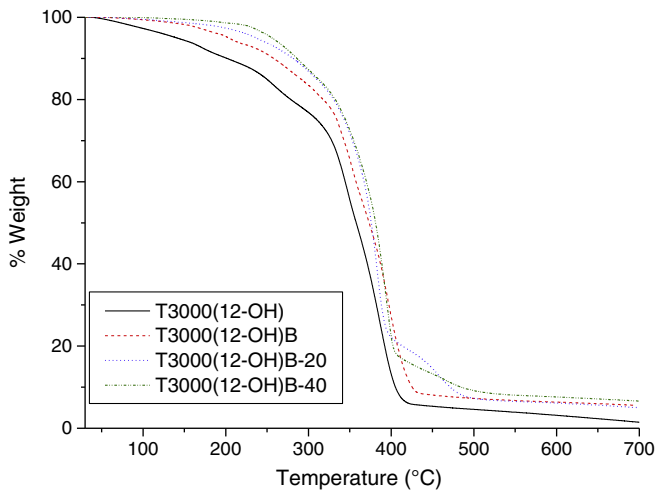


Fig. 6. Comparison of the weight loss of T3000(12-OH)B-Li with Li=20 and 40. The heating rate was 10 °C/min.

of the C–O–C peaks indicating a higher degree of interactions by co-ordination between Li^+ and PO [18]. The peak at 635 cm^{-1} belongs to $\text{CF}_3\text{SO}_3\text{Li}$. The peak at 1030 cm^{-1} can be attributed to stretching vibration of SO_3 unit of the dopant and increases with the dopant ratio [18].

3.3. Thermal properties

The neat dendrimer and the electrolytes were characterized by thermogravimetric analysis in order to determine the degradation temperatures (T_d 's).

The thermal degradation behaviors of the pure dendrimer T3000(12-OH), the boron containing dendrimer T3000(12-OH)B and the dendritic electrolytes T3000(12-OH)B-Li are depicted in Fig. 6. As seen in the figure, the onset of degradation temperature (T_d) of boron containing dendrimer T3000(12OH)B is approximately 150 °C. The initial decomposition temperature of boron containing crosslinked dendrimer is higher than that of pure dendrimer, which indicates that crosslinking treatment is effective to enhance the thermal stability. The doped boron containing dendrimer electrolytes thermally stable up to 250 °C and then the materials starts to decompose. The degradation temperature of the electrolytes increased with an increase of $\text{CF}_3\text{SO}_3\text{Li}$. The improvement in the degradation temperature is due to the fact that $\text{CF}_3\text{SO}_3\text{Li}$ is a reactive salt [19]. All the electrolyte samples can be said to be thermally stable up to 200 °C which is higher than the operating temperature of lithium ion batteries.

3.4. Ionic conductivity

A similar tendency in the frequency dependence of the conductivity was observed at all temperatures for all samples. Fig. 7 shows the frequency dependence of the ac conductivity (σ_{ac}) at various temperatures for dendritic electrolyte T500(6-OH)B-40. It can be observed that the ac conductivity versus frequency for various temperatures exhibits almost linear increasing at low frequencies (i.e., <10 Hz at 20 °C) which are assigned to the polarization of the blocking Pt electrodes, while well-developed frequency independent plateaus are observed at higher frequencies.

The switchover from the frequency independent to frequency dependent region marks the onset of conductivity relaxation that gradually shifts to higher frequencies as temperature increases (i.e., $>10^5 \text{ Hz}$ at 20 °C). The frequency independent conductivity is, in general, identified with the dc conductivity (σ_{dc}). The DC conductivities of the samples were derived from the plateaus of $\log \sigma_{ac}$ vs. $\log F$ by linear fitting of the frequency independent region of the σ_{ac} . A

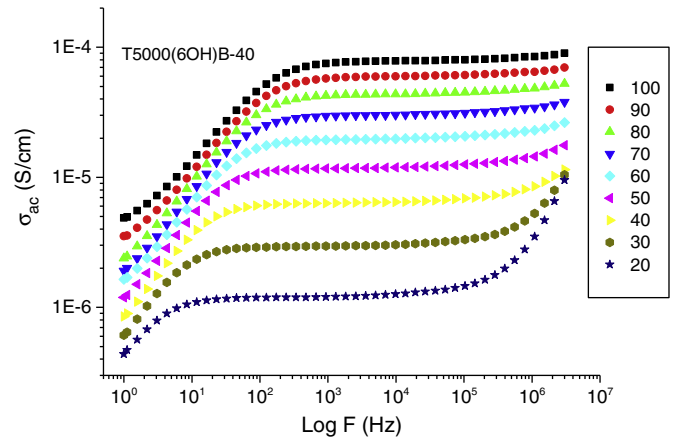


Fig. 7. AC conductivity versus frequency of anhydrous T5000(6OH)B-40 at various temperatures.

similar kind of frequency dispersion was observed for all other compositions as well.

The σ_{dc} versus temperature variations of the boron containing crosslinked dendritic electrolytes with different length of the core and concentration of LiCF_3SO_3 are shown in Fig. 8. It is obvious from the plots that the temperature dependence of ion conductivity did not follow the Arrhenius behavior. As expected the shapes of the conductivity curves were bent, indicating that the ion conduction in these crosslinked dendritic electrolytes can be explained by the concept of the free volume [20,21], which is expressed by the Vogel–Tamman–Fulcher (VTF) equation (Eq. (1)). To have better insight into the temperature dependence of σ_{dc} , the conductivity data have been fitted to the equation.

$$\log \sigma = \log \sigma_0 - Ev/[k(T - T_0)] \quad (1)$$

where σ_0 (S/cm) is the conductivity at infinite temperature, Ev is the Vogel activation energy (eV), k is the Boltzmann constant, T is the absolute temperature and T_0 is the Vogel temperature which is the start of the segmental relaxations (°C). The VTF parameters of the samples are given in Table 2.

As seen in Fig. 8, the conductivity isotherm indicates that the DC conductivity greatly depends on temperature, the doping ratio, core size and generation of dendrimers. The conductivity increases with the core size of dendrimer and decreases with a generation of the

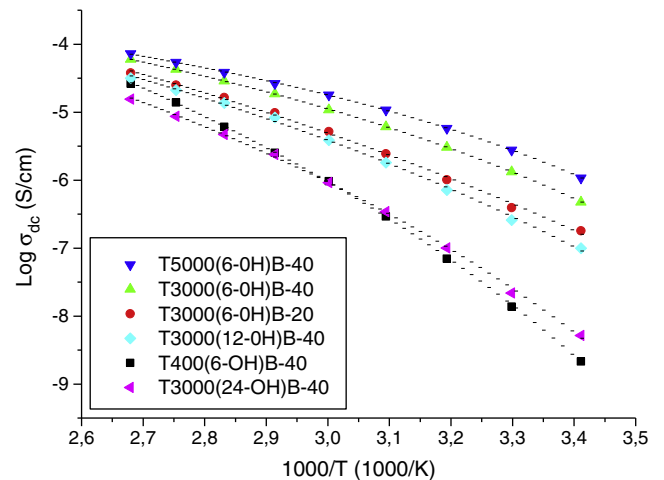


Fig. 8. Temperature dependence of the DC conductivities and fitted curves of TX(Y-OH) B-Li based materials.

Table 2E_(a), σ_o and T₀ values of the dendrimer electrolytes TA(XOH)B-Li.

	E _(a)	σ_o (S/cm)	T ₀ (°C)
T5000(6OH)-40	0.058	0.0047	−59.80
T3000(6OH)-40	0.084	0.0164	−72.97
T3000(6OH)-20	0.168	0.3876	−112.65
T3000(12OH)-40	0.143	0.1739	−95.10
T3000(12OH)-40	0.184	3.8979	−80.24
T5000(6OH)-40	0.151	0.3257	−77.37

dendrimer. The segmental motion of the dendritic electrolytes having small core size is suppressed by cross-linking of borate ester, because the electrolyte from the small core size of dendrimer has large amount of the cross-linking point (borate ester groups). At higher core size, the conductivity increases rapidly indicating that the core size can increase the dissociation of the lithium salt and reduce the aggregation of ions in the dendritic electrolytes.

Maximum ionic conductivity was measured for the dendritic electrolyte, T5000(6-OH)B-20 as $\sim 1.6 \times 10^{-4}$ S/cm at 100 °C. The conductivity increased gradually with increasing salt concentration and reached a maximum for PO/Li = 20. This maximum ion conductivity value may be explained by the increase in the concentration of charge carrier when salt is added to the dendrimer until a maximum value is reached, after which the ionic conductivity decreases [22].

Previously Dillon and Shriver prepared electrolytes by the introduction of Li[(CF₃SO₂)₂N] into amorphous branched polymers and dendritic macromolecular hosts [13]. In their study, amine dendrimers such as poly(propyleneimine) tetrahexacontaamine dendrimer (DAB-AM-64) and starburst poly(amidoamine) (PAMAM) were chosen because they do not contain rigid groups. These materials exhibit high ionic conductivity and become comparable to our system, TA(XOH)B-Li, that we report in this work.

4. Conclusion

In this work, various *cobweb* model boron-complexed dendritic electrolytes were prepared and characterized. These electrolytes were characterized using NMR, FT-IR, TGA and dielectric-impedance spectroscopy. The samples are thermally stable up to 200 °C. The conductivity studies illustrate that the ionic conductivity of these dendritic electrolytes was found to be dependent on core size and generation of dendrimer as well as the Li⁺ content. The dendritic electrolyte T5000(6-OH)B-20 illustrated a satisfactory ionic conductivity of 1.6×10^{-4} S/cm at 100 °C. The electrolytes with higher salt content show VTF behavior, indicating that the ion motion is depended on the flexibility of the dendrimer side chain. These

materials enable a safer, highly flexible, reproducible, and environmentally friendly for producing batteries with high ionic conductivity and good mechanical properties. These new electrolytes with these features can be suggested for application.

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